Here's the complete **reference material** for

1. **Setup and Install Docker**
2. **Docker Commands**

Each section includes explanations and examples

🛠 1. Setup and Install Docker

✅ Installing Docker on Windows

Steps:

1. Download Docker Desktop from:
   👉 https://www.docker.com/products/docker-desktop
2. System Requirements:
   - Windows 10/11 Pro, Enterprise, or Education
   - WSL 2 (Windows Subsystem for Linux) enabled
3. Installation:
   - Run installer and follow prompts
   - Choose WSL 2 as the backend
   - Restart system if prompted
4. Verify:
5. docker --version

🔎 Windows-specific Considerations:

- Enable **virtualization** in BIOS
- Enable WSL 2 and **Hyper-V**
- To fix permission issues:
- net localgroup docker-users <your-username> /add

🍎 Installing Docker on Mac

Steps:

1. Download Docker Desktop for Mac from the official site
2. System Requirements:
   - macOS 11+ (Intel or Apple Silicon)
3. Install:
   - Drag Docker.app to Applications

- Start Docker from Launchpad

4. Verify:
5. docker --version

🐧 Installing Docker on Linux

✅ Ubuntu:

sudo apt update

sudo apt install docker.io

sudo systemctl start docker

sudo systemctl enable docker

✅ CentOS:

sudo yum install -y yum-utils

sudo yum-config-manager --add-repo \

https://download.docker.com/linux/centos/docker-ce.repo

sudo yum install docker-ce docker-ce-cli containerd.io

sudo systemctl start docker

sudo systemctl enable docker

🔧 Post-installation Steps

1. Run Docker without sudo:
2. sudo usermod -aG docker $USER

Then log out and log back in.

3. Test installation:
4. docker run hello-world

🛠 Troubleshooting Common Issues

Problem	Solution
Docker won't start	Restart Docker Desktop or system
WSL2 backend error	Ensure WSL2 kernel update installed
Permission denied	Add user to docker-users or docker group
Containers not running	Check virtualization is enabled in BIOS

▯ 2. Docker Commands

▯ Basic Docker Commands Overview

Command	Description
docker run	Run a container
docker ps	List running containers
docker stop <id>	Stop a container
docker rm <id>	Remove a container
docker images	List local images
docker rmi <id>	Remove an image
docker build -t name .	Build image from Dockerfile
docker pull <image>	Download image
docker push <image>	Upload image to Docker Hub

🔧 Practical Examples

✅ **Running Hello World:**

```
docker run hello-world
```

✅ **Run Ubuntu in Interactive Mode:**

```
docker run -it ubuntu /bin/bash
```

✅ **Run NGINX in Detached Mode with Port Mapping:**

```
docker run -d -p 8080:80 nginx
```

✅ **View Running Containers:**

```
docker ps
```

✅ **View All Containers:**

```
docker ps -a
```

✅ **Stop & Remove Containers:**

```
docker stop <container_id>
docker rm <container_id>
```

✅ **Build Image from Dockerfile:**

docker build -t myimage .

✅ **Pull and Push Image:**

docker pull python:3.10

docker push your-dockerhub-username/myimage

🏃 **Docker Run: Modes**

1. Interactive Mode

Runs a container where you can interact with the shell.

docker run -it ubuntu /bin/bash

- -i: Interactive
- -t: Allocate pseudo-TTY (terminal)

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2. Detached Mode

Runs the container in the background.

docker run -d nginx

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3. Port Mapping

Maps host port to container port.

docker run -d -p 8080:80 nginx

- Access: http://localhost:8080